Jenkins Setup and Configuration Guide

IP Management

- **Dynamic IP Change:** When you stop your instance, the server's IP may change. To avoid disruptions, consider using an **Elastic IP**. You can allocate an Elastic IP and assign it to your instance for consistent access.
- **Updating Jenkins URL:** After allocating an Elastic IP, update the Jenkins URL in the configuration settings to ensure accurate access.

Jenkins Service Management

- **Starting Jenkins:**

To start Jenkins via CLI, use:

```
sudo systemctl start jenkins
```

    - Alternatively, you can restart Jenkins through the GUI by navigating to the URL `/restart`.

Safe Restart: If you need to restart Jenkins while jobs are running, use:

```
/safeRestart
```

- This allows Jenkins to wait for currently running jobs to finish before restarting, preventing interruptions.

Build Jobs and Logs

- **Viewing Build Logs:** After job completion, logs are accessible in the build workspace. You can navigate to:
    - `cd <workspace>` (e.g., `cd $HOME/<workspace>`)
    - View build history and logs to diagnose issues or verify builds.

Job Creation and Configuration

- **Discard Old Builds:** When creating jobs, configure the job to discard old builds after a specified number to save space.
- **Parameterized Builds:** You can set up parameterized jobs to deploy to different environments (e.g., UAT, Lab, Sandbox) based on specified parameters.
- **Throttle Builds:** To manage system load, use the Throttle Builds plugin to limit the number of builds executed concurrently over a specified time period.
- **Custom Workspace:** Create a custom workspace for each job to organize project files and build artifacts efficiently.

Source Code Management

- **Git Integration:**
  - Add your source code repository URL and credentials in the job configuration.
  - Configure build triggers to automatically trigger builds upon code changes in the Git repository.

Build Scheduling

- **Cron Jobs for Automation:**
  - Use cron expressions to schedule periodic builds in the job configuration.

Example cron format:

```
* * * * *
```

- The first star represents the minute, the second for the hour, the third for the day of the month, the fourth for the month, and the fifth for the day of the week. You can find detailed cron syntax help online.

Build Steps

- **Maven Configuration:**
  - Install Maven on your server.
  - Create a `pom.xml` file to specify the project details and dependencies.
  - Use Maven to build your project and generate artifacts (e.g., `.war` or `.jar` files).
- **Deploying to Tomcat:**
  - Deploy the generated artifacts to your Tomcat server.
  - Before install tomcat.
  - Install necessary plugins in Jenkins for enhanced deployment functionality (e.g., to deploy directly to a container).
- **Tomcat Configuration:**
  - Provide your Tomcat credentials and URL in the Jenkins job configuration to facilitate deployment.
  - After setting up, trigger a build to deploy the application to the server.

Final Steps

- **Verification:** After deployment, access your application through the web to ensure everything is functioning correctly.